

SUNIL HANSDA

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PROFILE SUMMARY

Software Engineer with 3 years of experience developing **Python-based backend services, workflow automation platforms, and AI-enabled solutions** using **Python, Pandas, FastAPI, SQL, AsyncIO, and distributed systems**. Experienced in automating metadata-driven ETL workflows, data ingestion and processing, pipeline orchestration, execution monitoring, and failure recovery across **200+ enterprise pipelines**. Hands-on experience building **Agentic AI workflows using LangGraph and custom tools**, with working knowledge of RAG and MCP-based integration patterns. Skilled in **SQL Server, PostgreSQL, MySQL, MongoDB, Redis, Kafka, Docker, Kubernetes, GCP, Azure, and GitLab CI/CD**, with a strong foundation in automated testing, debugging, platform reliability, and cloud-native engineering.

SKILLS

Programming & Data Processing: Python, Pandas, SQL, Shell Scripting, Data Structures, Algorithms, OOP, Design Patterns

Data Engineering: ETL Pipelines, Data Ingestion, Data Transformation, Data Validation, Data Quality, Metadata Management, Workflow Orchestration, Batch Processing

Backend Engineering: FastAPI, Flask, REST APIs, AsyncIO, Microservices, Distributed Systems, API Design, JWT Authentication

Databases & Messaging: SQL Server, PostgreSQL, MySQL, MongoDB, Redis, Kafka, Relational and NoSQL Databases

AI & Agentic Systems: Agentic AI, AI Agents, LangGraph, LangChain, Model Context Protocol (MCP), Custom Tools, RAG, Azure OpenAI, Prompt Engineering, Context Grounding, AI Output Validation

Testing & Monitoring: Pytest, Unit Testing, Integration Testing, API Testing, Data Validation, Structured Logging, Execution Monitoring, Root-Cause Analysis, Performance Optimization

Cloud & DevOps: GCP, Azure, Docker, Kubernetes, Git, GitLab CI/CD, Linux, HashiCorp Vault, Secrets Management

WORK EXPERIENCE

CGI Inc., Bangalore

July 2023 – Present

Software Engineer

AI-Enabled Workflow Automation & Platform Engineering

- Developed and maintained production-grade workflow automation and data-processing services using **Python, Pandas, FastAPI, SQL, and AsyncIO** for enterprise ETL migration and execution workflows.
- Built **15+ asynchronous REST APIs** for data ingestion, validation, transformation, workflow orchestration, execution tracking, metadata processing, and enterprise-system integration.
- Designed reusable, metadata-driven modules to orchestrate and monitor **200+ ETL pipelines**, reducing manual pipeline configuration effort by **40%**.
- Implemented data validation, schema checks, exception handling, retries, failure recovery, structured logging, and execution monitoring to improve pipeline reliability and data quality.
- Integrated data services with **SQL Server, Redis, and Kafka** for metadata storage, caching, workload coordination, and event-driven processing.
- Optimized SQL queries, indexes, and data-access paths, improving metadata retrieval and pipeline-processing performance by **30%**.
- Developed automated operational reports and monitoring views for pipeline status, execution failures, processing metrics, and deployment validation.
- Built Agentic AI workflows using **LangGraph, custom tools, context grounding, and output validation** to automate workflow configuration and execution decisions.
- Applied **MCP concepts** for standardized AI tool and enterprise data-source integration while evaluating reusable patterns for production workflows.
- Created **Pytest** unit, integration, API, data-validation, and failure-scenario tests and participated in code reviews, production debugging, and root-cause analysis.
- Tech Stack:* Python, Pandas, FastAPI, AsyncIO, Pytest, SQL Server, Redis, Kafka, LangGraph, Docker, GCP

Cloud Platform, DevOps & Systems Reliability

- Automated validation, configuration, packaging, and deployment of **1,200+ ETL and platform artifacts**, reducing deployment effort by **50%**.
- Developed reusable GitLab CI/CD modules for automated testing, data and configuration validation, release packaging, and multi-environment deployments.
- Supported containerized data-processing services and orchestration workflows using **Docker, Kubernetes, GCP, Linux, and GitLab Runners**.
- Integrated **HashiCorp Vault** with CI/CD pipelines to securely manage credentials and remove sensitive information from source code and deployment configurations.
- Performed pipeline monitoring, root-cause analysis, performance tuning, deployment troubleshooting, and production support in collaboration with engineering, data, product, and Agile teams.

PROJECT WORK

GenAI Medical Assistant – CGI Internal Hackathon

June 2025

- Built a containerized AI application using **Python, FastAPI, Azure OpenAI, and Angular**, exposing HTTP services for symptom input, retrieval, and context-grounded response generation.
- Implemented a RAG ingestion and retrieval pipeline with document parsing, embeddings, **pgvector** hybrid search, reranking, and semantic guardrails to reduce hallucinations.
- Created a **Pytest evaluation harness** for retrieval recall, response accuracy, safety compliance, and regression testing.
- Designed modular AI-agent components and tool-integration boundaries aligned with modern Agentic AI and MCP-based platform patterns.
- Tech Stack:* Python, FastAPI, Azure OpenAI, PostgreSQL/pgvector, LangChain, RAG, Pytest, Angular, Docker

EDUCATION

National Institute of Technology, Jamshedpur

July 2019 – June 2023

B.Tech in Electrical Engineering

CGPA: 8.12/10

AWARDS AND CERTIFICATES

- Databricks Certified Generative AI Engineer:** Vector Search, Model Serving, MLflow, Unity Catalog, RAG systems, and LLM orchestration.
- Bronze Award (CGI):** Recognized for metadata-driven deployment automation that reduced manual operations by 50%.